MILITARY SPECIFICATION

SIGNALS, ILLUMINATION AIRCRAFT DOUBLE-STAR AN-M37A2 THROUGH AN-M42A2 (ALTERNATIVE CASE) PARTS, AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts and the loading, assembling and packing for one type of aircraft double-star signals with plastic noryl case/top system designated as follows:

AN-M37A2 - Red Stars
AN-M38A2 - Yellow Stars
AN-M39A2 - Green Stars
AN-M40A2 - Red and Yellow Stars
AN-M41A2 - Red and Green Stars
AN-M42A2 - Green and Yellow Stars

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223 - Powder, Black
MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

FSC: 1370

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-286 - Propellants; Solid, Sampling, Examination, and Testing
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY MUNITIONS COMMAND

8836949 - Cartons Packing, Ammunition, For Signals, Illumination, Aircraft, and Simulator, Projectile, Air Burst.
8836950 - Box, Packing, Ammunition, For Signals, Illumination, Aircraft, and Simulator, Projectile, Air Burst.
9201136 - Tunnel, Light
9201268 - Procedure, Light Output Measurement
9201390 - Procedure, Photocell Checkout
9201392 - Procedure, Color Value Measurement
9278328 - Signal, Illumination, Aircraft, Double-Star, AN-M37A2 through AN-M42A2 Assemblies (Alternative Case).

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material .-Materials shall be in accordance with applicable drawings and specifications.

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3.2 Assembly.-The assembly shall comply with all requirements specified on Drawing (dwg.) 9278328 and with all requirements specified in applicable specifications.

3.3 Moisture content

3.3.1 Star charge composition.-The moisture content of the red star charge composition at the loading station at the time of loading shall not exceed 0.1 percent and the moisture content of the green and yellow star charge compositions at the loading station at the time of loading shall not exceed 0.5 percent.

3.3.2 Black powder.-The moisture content of the black powder at the loading station at the time of loading the signals, shall not exceed 0.3 percent.

3.3.3 Paper, felt and chipboard.-The moisture content of the paper, felt and chipboard components, at the loading station at the time of loading the signals shall not exceed 6 percent.

3.3.4 Polypropylene felt.-The moisture content of the polypropylene felt components, at the loading station at the time of loading shall not exceed 0.1 percent.

3.4 Transportation vibration.-The signal assemblies shall comply with the following requirements when tested as specified in MIL-STD-331, Test 119, Procedure 2.

3.4.1 The signal assembly shall not function during the test.

3.4.2 The signal assembly shall be safe to transport following the test (see 6.6).

3.4.3 There shall be no evidence of external damage to the signal assemblies that will affect the intended function (see 6.7).

3.4.4 There shall be no evidence of damage in the packing or packaging that would affect the storage life of the signal assemblies.

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3.4.5 The signal assembly shall comply with the requirements of 3.5 following this test.

3.5 Functioning (Proving Ground) .-The signals shall function in accordance with the following requirements when tested.

3.5.1 No star shall burst in the pistol.

3.5.2 The signal shall not split or rupture; nor shall It bulge to the extent that it sticks or “freezes” in the pistol barrel.

3.5.3 The primer shall not fail to function.

3.5.4 The burning stars shall not fail to attain a minimum (min.) altitude of 200 feet above the firing point.

3.5.5 The stars shall not fail to ignite and reach approximately 1/4 of full brilliance within 2 seconds after firing.

3.5.6 Portions of composition that become detached during star burning shall not burn for more than two seconds.

3.6 Air leakage test .-The signal shall show no evidence of leakage when tested as specified.

3.7 Jolt test .-The signal assemblies shall comply with the following requirements when subjected to the jolt test:

3.7.1 The signal assembly shall not function during test.

3.7.2 There shall be no detachment of the top, loose or protruding primers or other external damage to the signal. (See 6.7).

3.7.3 The signal assembly shall be safe to handle after test. (See 6.6.1).

3.8 Workmanship. -All parts shall be fabricated and loaded In a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust, crazes, unblended radii, porosity, warpage, burn marks, checks, chipped edges, blisters, excess flash, grease, salt deposits, corrosion products, and other defects and foreign matter which would affect their serviceability. The

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cleaning method used shall not be injurious to any part nor shall the. parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

3.9 First article inspection, -This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection **re-**quirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product .-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate svbmitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority till not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 15 complete sets of parts and sub-assemblies (i.e. 15 each of every component and subassembly, including at least three (3) items from each cavity used to produce plastic parts and be so identified as to permit tracing each item to the cavity in which it was molded); and ten (10) star assemblies of each color; and forty (40) signal assemblies of each color designation which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials Including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall Inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including a certificate of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse In production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 First article inspection provisions

4.2.2.1 Inert parts

4.2.2.1.1 Submission.-The 15 sets of inert parts (see 4.2.1)

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shall be submitted to a Government approved laboratory designated by the contracting officer.

4.2.2.1.2 Inspections to be performed.-The inert parts will be subjected to any or all of the applicable examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all of the requirements of the applicable drawings.

4.2.2.1.3 Rejection.-If any component or subassembly fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an inert component or subassembly to comply with any of the stated requirements.

4.2.2.2 Star assemblies

4.2.2.2.1 Submission.-The 10 star assemblies (see 4.2.1) shall be submitted to a Government approved laboratory designated by the contracting officer.

4.2.2.2.2 Inspection to be performed.-The star assemblies shall be subjected to the static test for candlepower, color value and burning time in accordance with the method specified in 4.4.5.

4.2.2.2.3 Rejection.-The first article sample shall be *re*-jected if any star assembly fails to comply with any of the requirements as classified in 4.3.3.5, Table IV. The Government reserves the right to terminate Its inspection upon encountering any failure of an item in the sample to comply with any or all requirements.

4.2.2.3 Loaded signal assemblies

4.2.2.3.1 Submission.-Thirty (30) loaded signal assemblies of each designation (from 4.2.1) shall be submitted to a Government proving ground designated by the contracting officer.

4.2.2.3.2 Inspections to be performed.-Prior to submission of the signal assemblies to the proving ground for functioning test, the loaded components shall be subjected to the following tests at the contractor’s plant by the Government:

a. Transportation vibration (see 3.4) - 30 signal assemblies of each designation.

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b. Air leakage (see 3.6) - 30 signal assemblies of one designation selected from the assemblies after test specified in "a" above. After completion of the transportation vibration and air leakage tests, each signal assembly shall be conditioned and fired at the proving ground as specified in 4.4.3.

4.2.2.3.3 Rejection. -The first article sample shall be rejected If any loaded signal assembly fails to comply with:

a. The requirements of 3.4 as classified in 4.3.3.2, Table 11.

b. The requirements of 3.6 as classified in 4.3.3.4.

The requirements of 3.5 as classified in 4.3.3.3, Table III.

The Government reserves the right to terminate its inspection upon encountering any failure of an Item in the sample to comply with any or all requirements.

4.2.2.4 Jolt test

4.2.2.4.1 Submission.-Ten (10) loaded signal assemblies (from 4.2.1) shall be submitted to a Government approved laboratory designated by the Contracting Officer.

4.2.2.4.2 Inspections to be performed ten (10) signal assemblies shall be subjected to the jolt test as specified in 4.4.7.

4.2.2.4.3 Rejection. -The first article sample shall be rejected if any signal assembly fails to comply with the requirements of 3.7 as classified in 4.3.3.8, Table V. The Government reserves the right to terminate its Inspection upon encountering any failure of an item *in* the sample to comply with any or all requirements.

4.3 Inspection provisions

4.3.1 Lot formation. -The term "inspection lot" as used in this specification is defined as an essentially homogeneous col-

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lection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which Is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded signals shall contain:

a. Primers of one lot interfix number from one manufacturer.

b. Black powder from not more than **one lot.**

c. Signals of one designation only.

d. Star charge composition produced by one manufacturer under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.3.2 Examination. -The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for

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each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable sub-paragraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Case, Signal (See dwg. 9224033 covering a detail 278328).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:			
1.	Crack or split	Visual	01001
Major:	AQL 1.00 percent		
101.	Diameter of primer hole, maximum (max.)	Gage	01002
102.	True position of primer hole with flange	Gage	01003
103.	True position of primer hole counterbore with primer hole	Gage	01004
104.	Runout, outside diameter with inside diameter at 0.25 basic length and inside diameter at 2.75 basic length	Gage	01005
105.	Minimum (min.) wall thickness at open end	Gage	01006
106.	Length of flange	Gage	01007
107.	Pit marks, sinks or other surface imperfection	Visual	01008
Minor:	AQL 2.50 percent		
201.	Diameter of flange	Gage	01009
202.	Angle of flange	Gage	01010
203.	Length from top of powder cavity to base	Gage	01011
204.	Total length	Gage	01012
205.	Discoloration of case	Visual	01013
206.	Evidence of poor workmanship (see 3.8)	Visual	01014

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4.3.2.2 Top (see dwg. 9224028 covering a detail of 9278328).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Energy director missing or not full 360°	Visual	02001
102.	Height of energy director	Gage	02002
103.	Concentricity of outside diameter at shoulder with large outside diameter	Gage	02003
Minor:	AQL 1.50 percent		
201.	Outside diameter	Gage	02004
202.	Height	Gage	02005
203.	Thickness	Gage	02006
204.	Evidence of poor workmanship (see 3.8)	Visual	02007

4.3.2.3 Star assembly (see dwg. 8847461 covering a detail of dwg. 9278328).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color identification incorrect	Visual	03001
Major:	AQL 0.65 percent		
101.	Length of protruding quickmatch	Gage	03002
102.	Priming charge distribution of total hole area, less than 2/3 min	Visual	03003
103.	Quickmatch missing	Visual	03004
104.	Disc loose	Manual	03005
Minor:	AQL 1.00 percent		
201.	Crimp not full 360 degrees	Visual	03006
202.	Evidence of poor workmanship (see 3.8)	Visual	03007

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4.3.2.4 Assembly, prior to inserting star assemblies, retaining washer and wad-disc assembly (see dwg. 9278328).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Propellant charge missing.....	Visual	04001
2.	Propelling charge weight less than min. (see Note 1).....	Balance	04002
Major:	AQL 0.40 percent		
101.	Primer above flush or greater than max below flush.....	Gage	04003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8).....	Visual	04004

NOTE 1: Volumetric weighing may be utilized as a substitute for precise weighing following qualification. Qualification shall consist of precise weighing until such time as twenty five hundred (2,500) consecutively volumetric loaded charges are found to meet the minimum prescribed weight.

After successful completion of the qualification quantity, the contractor may use a 32-0-1 check weighing sampling plan for each hours production of powder loaded signal cases. If a "weight under min." critical defect is found in the thirty two (32) unit sample, the hours production represented by the sample plus enough additional units to total eight hundred and forty (840) consecutive, defect free units, must be produced and check weighed before returning to the 32-0-1 hourly sampling plan.

4.3.2.5 Assembly, prior to inserting star assemblies (see dwg. 9278328).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Wad-disc assembly missing, not seated, or disc punctured.....	Visual-Manual	05001

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Major: None defined

Minor:			
	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8).....	Visual	05002

4.3.2.6 Assembly, prior to inserting top (see 9278328).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color combination of star assemblies incorrect.....	Visual	06001
Major:	AQL 0.40 percent		
101.	Star assembly missing, inverted or damaged to the extent that function may be impaired.....	Visual	06002
102.	Cushioning wad or spacer disc missing.....	Visual	06003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8).....	Visual	06004

4.3.2.7 Signal assembly (see 9278328).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Band color incorrect.....	Visual	07001
Major:	AQL 1.00 percent		
101.	Top insecure or not fully seated.....	Manual	07002
102.	Contents loose.....	Manual	07003
103.	Color band missing.....	Visual	07004
104.	Marking misleading or unidentifiable.....	Visual	07005
105.	Case split or assembly otherwise damaged.....	Visual	07006

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Minor:	AQL 1.50 percent		
201.	Total length.....	Gage	07007
202.	Bare spot in protective coating....	Visual	07008
203.	Finish improper.....	Visual	07009
204.	Evidence of poor workmanship (see 3.8).....	Visual	07010

4.3.2.8 Carton, prior to sealing (see 8836949).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Number of assemblies in carton incorrect.....	Visual	08001
102.	Assembly improperly packed in carton (inverted or packing material missing).....	Visual	08002
Minor:	None defined		

4.3.2.9 Sealed carton (see dwg 8836949).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color identification missing or incorrect.....	Visual	09001
Major:	AQL 0.40 percent		
101.	Sealing compound inadequate.....	Visual	09002
102.	Carton damaged to extent that contents are exposed or liable to become exposed.....	Visual	09003
Minor:	AQL 1.00 percent		
201.	Contents loose.....	Manual	09004
202.	Marking, other than color identification, misleading or unidentifiable.....	Visual	09005

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4.3.2.10 Wood packing box, prior to sealing (see dwg. 8836950).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Carton inverted (primer end up)....	Visual	10001
Minor:	AQL: 0.65 percent		
201.	Incorrect number of cartons.....	Visual	10002

4.3.2.11 Sealed wood packing box (see dwg. 8836950).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color identification missing or incorrect.....	Visual	11001
Major:	AQL 0.65 percent		
101.	Box damaged to extent that contents are exposed or liable to become exposed.....	Visual	11002
102.	Hardware or strapping missing broken or loose.....	Visual-Manual	11003
103.	DOD symbol misleading or unidentifiable.....	Visual	11004
Minor:	AQL 2.50 percent		
201.	Hardware or strapping improperly engaged or assembled.....	Visual-Manual	11005
202.	Car seal missing, unsealed, or improperly positioned.....	Visual	11006
203.	Contents loose.....	Manual	11007
204.	Marking other than color identification or DOD symbol misleading or unidentifiable.....	Visual	11008

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4.3.2.12 Workmanship.-Applies to all components and assemblies not specifically listed in 4.3.2.1 through 4.3.2.11 (see cable drawing).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:	AQL 0.40 percent per defect	
101.	Operation missing or improperly performed	Visual
102.	Burr or chip on or at intersection of sliding or other functional surfaces	Visual-Manual
103.	Finish of sliding or other functional surface improper	Visual
104.	Component missing or improperly assembled	Visual
105.	Protective finish missing or contains bare spot	Visual
Minor:	AQL 0.65 percent	
201.	Evidence of poor workmanship other than that covered under Major category above (see 3.8)	Visual

4.3.3 Production testing.-Unless otherwise specified, the following tests shall be performed using Standard MIL-STD-105 for sampling plan requirements.

4.3.3.1 Moisture content (see Table I) Major defects.

TABLE I

Material	Code No.
Star charge composition (see 3.3.1)	12001
Black powder (see 3.3.2)	12002
Felt (see 3.3.4)	12003
Paper (see 3.3.3)	12004
Chipboard (see 3.3.3)	12005
Felt, Wool (See 3.3.3)	12006

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The contractor shall provide adequate controls to insure that the materials comply with the requirements. For verification, the contractor shall select and test one sample of each material used in each eight hours production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirement and loading has not begun, the material represented by the sample shall be rejected. If components have been loaded with material containing excessive moisture, the remaining unloaded material and loaded components shall be rejected. Test procedures shall be as specified in 4.4.1. The sample of star charge composition shall be selected prior to consolidating the star assemblies.

4.3.3.2 Transportation vibration.-The signal assemblies shall be observed and examined visually, without disassembly, for any evidence of failure to comply with the requirements as classified in Table II.

TABLE II

Defect	Classification	Code No.
Signal functions during test (see 3.4.1)	Critical	07011
Signal not safe to transport following test (see 3.4.2)	Critical	07012
Signal damaged after test (see 3.4.3)	Major	07013
Signal packing damaged (see 3.4.4)	Major	07014

4.3.3.2.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria of 4.3.3.2.1 and 4.3.3.3.1, two hundred anti forty (240) signal assemblies shall be selected from each lot for test in accordance with the procedure specified In 4.4.2. The lot shall be rejected if any defect classified In Table II occurs.

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4.3.3.2.2 After three consecutive lots have complied with the acceptance criteria of 4.3.3.2.1, eighty (80) signal assemblies shall be selected from each lot for test. The lot shall be rejected if any defect classified in Table II occurs.

4.3.3.3 Functioning. -The signal assemblies shall be observed for any evidence of failure to comply with the requirements as classified in Table III when tested as specified in 4.4.3.

TABLE III

Defect	Classification	Code No.
Star color incorrect (see dwg. 8847461)	Critical	07015
Star bursts in pistol (see 3.5.1)	Critical	07016
One star fails to ignite (see 3.5.5)	Critical	07017
Case splits, ruptures or sticks in barrel (see 3.5.2)	Major	07018
Primer fails to function (see 3.5.3)	Major	07019
Star intensity and ignition time over max. (see 3.5.5)	Major	07020
Detached composition burning time over max. (see 3.5.4)	Major	07021
Altitude below min. (see 3.5.4)	Major	07022
Star burning time below min. (see dwg. 8847461)	Major	07023

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4.3.3.3.1 Two hundred (200) of the signal assemblies selected and tested in accordance with 4.3.3.2.1 shall be forwarded to the Proving Ground for this test. The lot shall be rejected if a Critical Defect occurs or if eight (8) or more Major defectives are found during the test (see Table III).

4.3.3.3.2 The signal assemblies selected and tested in accordance with 4.3.3.2.2 shall be forwarded to the Proving Ground for this test. The lot shall be rejected if a Critical Defect occurs or If four (4) or more Major defective are found during the test (see Table III).

4.3.3.4 Air leakage test (see 3.6) Major defect - Code No. 07024.-This test shall be performed 100 percent in accordance with the procedure specified in 4.4.4. All assemblies that fail to comply with the requirement shall be classed defective and removed from the lot.

4.3.3.5 Static test of star assembly (see dwg. 8847461).- The star assemblies shall be tested and observed for the defects as classified in Table IV.

TABLE IV

Defect	Classification	Code No.
Assembly fails to ignite	Major	03008
Candlepower under min.	Major	03009
Color value under min.	Major	03010

4.3.3.5.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, eighty (80) star assemblies of each color shall be selected from each lot for test in accordance with the procedure specified in 4.4.5. The lot shall be rejected if four (4) or more defective are found during the test.

4.3.3.5.2 After three consecutive lots have complied with the acceptance criteria of 4.3.3.5.1, thirty two (32) star assemblies of each color shall be selected from each lot for test. The lot shall be rejected, if, during the test, three (3) or more assemblies exhibit any of the defects as listed in Table IV.

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4.3.3.6 Chamber gaging (see dwg. 9278328) Major defect - Code No. 07025.-Chamber gaging shall be performed 100 percent. Any assembly that binds, or otherwise fails to gage shall be classed defective and removed from the lot. The gaging shall be performed as specified in 4.4.6.

4.3.3.7 Check test for possible deterioration of primers (see applicable primer specification. -If the total elapsed time between original acceptance of any primer lot and the assembly of that lot into the signal assemblies exceeds two years, or If the primers have been subjected to adverse conditions, however, brief, at any time since previous tests, the primer lots shall be subjected to and must satisfactorily pass the check test specified in the applicable primer specification. The check test shall be performed by the contractor (see 6.5) prior to assembling the primers into the signals.

4.3.3.8 Jolt test (see 3.7).-The signal assemblies shall be observed and examined visually without disassembly for evidence of failure to comply with the requirements as classified in Table V.

TABLE V

Defect	Classification	Code No.
Signal functions during test (see 3.7.1)	Critical	13001
Signal top loose or detached from case (see 3.7.2)	Major	13002
Signal assembly damaged, after test (see 3.7.2)	Major	13003
Signal not safe to handle after test (see 3.7.3)	Critical	13004

4.3.3.8.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the requirements, thirty two (32) signal assemblies shall be

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selected from each lot for test in accordance with the procedure specified in 4.4.7. The lot shall be rejected if any defect classified in Table V occurs.

4.3.3.8.2 After three (3) consecutive lots have complied with the criteria of 4.3.3.8.1, twenty (20) signal assemblies shall be selected from each lot for test in accordance with the procedure specified in 4.4.7. The lot shall be rejected if any defect classified in Table V occurs.

4.3.4 Inspection Equipment .-Inspection equipment required to perform the Inspections prescribed in this specification is identified in the Examinations (4.3.2 and 4.3.5) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection design and approval.

4.3.5 Dimensional control of molded parts.-In place of normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (see Note 2) a sample of three (3) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.3.5.1 through 4.3.5.2 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the contracting officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-A-M for approval prior to manufacture

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of equipment. The noted subparagraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 2. - In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. reins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Picatinny Arsenal, Dover, New Jersey 07801 ATTN: SARPA-QA-A-M. If any new change in material, cycle time, a reworked cavity, or a new cavity is used, a new curing time shall be established and approved.

4.3.5.1 Case Signal (see dwg. 9224033 covering a detail of 9278328).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Crack or split.....	Visual	01001
Major:			
101.	Diameter of primer hole, max.....	Gage	01002
102.	True position of primer hole with flange.....	Gage	01003
103.	True position of primer hole counterbore with primer hole.....	Gage	01004
104.	Runout, outside diameter with inside diameter at 0.25 basic length and inside diameter at 2.75 basic length.....	Gage	01005
105.	Min wall thickness at open end....	Gage	01006
106.	Inside diameter at basic length near mouth.....	Gage	01015
107.	Inside diameter at basic length near base.....	Gage	01016
108.	Outside diameter at basic length near mouth.....	Gage	01017
109.	Outside diameter at basic length near base.....	Gage	01018

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110.	Length of flange	Gage	01007
111.	Pit marks, sinks or other surface imperfections	Visual	01008
112.	Depth of primer hole counterbore	Gage	01019
Minor:			
201.	Diameter of flange	Gage	01009
202.	Angle of flange	Gage	01010
203.	Thickness of base at primer hole	Gage	01020
204.	Total length	Gage	01012
205.	Diameter of powder cavity	Gage	01021
206.	Radius at base of powder cavity, max	Gage	01022
207.	Radius on outside of powder cavity wall	Gage	01023
208.	Inside diameter of mold relief core	Gage	01024
209.	Outside diameter of mold relief core	Gage	01025
210.	Diameter of primer housing boss	Gage	01026
211.	Radius of primer housing boss, max	Gage	01027
212.	Depth of mold relief core	Gage	01028
213.	Length from top of powder cavity to base	Gage	01011
214.	Discoloration of case	Visual	01013
215.	Evidence of poor workmanship (see 3.8)	Visual	01014

4.3.5.2 Top (see dwg. 9224028 covering a detail of 9278328).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Energy director missing or not full 360°	Visual	02001
102.	Height of energy director	Gage	02002
103.	Concentricity of outside diameter at shoulder with large outside diameter	Gage	02003

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104.	Thickness of flange.....	Gage	02008
105.	Thickness of energy director.....	Gage	02009
106.	Distance from edge to energy director.....	Gage	02010
107.	Diameter of energy director.....	Gage	02011
108.	Diameter at shoulder.....	Gage	02012
Minor:			
201.	Outside diameter.....	Gage	02004
202.	Height.....	Gage	02005
203.	Thickness.....	Gage	02006
204.	Inside diameter.....	Gage	02013
205.	Height of center raised portion...	Gage	02014
206.	Inside radius.....	Gage	02015
207.	Evidence of poor workmanship (see 3.8).....	Visual	02007

4.4 Test methods and procedures

4.4.1 Moisture content

4.4.1.1 Star charge composition

4.4.1.1.1 Preferred method.-The Karl Fisher method as stated in MIL-STD-1234, Method 101.2 up to paragraph 5.3 shall be used. A fifty gram sample plus or minus 0.1 gram shall be added to a 500 milliliter (ml) volumetric flask containing approximately 300 to 400 ml. of methanol and 25 grams of dry sodium nitrate. The flask shall be stoppered and the contents swirled cautiously for several minutes until the material is thoroughly dispersed. The sample shall be allowed to remain in contact with the methanol for approximately two hours. Then the 500 m. volumetric flask shall be filled up to the 500 ml. mark with methanol and swirled again. A blank without the sample shall be put through the same procedure. After the sample has settled, a 50 ml. aliquot of the clear supernatant liquid shall be withdrawn and, immediately, transferred to the standard titration vessel containing approximately 100 ml. of methanol which has just been titrated to the preliminary end point as described in MIL-STD-1234, Method 101.2, paragraph 5.1. The final end point shall be reached in 3.5 minutes in the manner described in MIL-STD-1234. Method 101.2 paragraph 5.4. A 50 ml. aliquot of the blank shall be titrated in the same manner. The water content shall be calculated as follows:

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Percent water equal (VR-S) - (V'R-S') Multiply the result by 100F and divide by W.

where:

F = grams of water per ml. of standard water in methanol solution.

V = ml. of Karl Fischer reagent added to the sample.

V' = ml. of Karl Fischer reagent added to the blank.

R = ml. of standard water in methanol solution per ml. of Karl Fischer reagent.

S = ml. of standard water in methanol solution for titration of sample.

S' = ml. of standard water in methanol solution for back titration of blank.

W = weight of sample in grams.

4.4.1.1.2 Alternate method.-The moisture content shall be determined in accordance with Method T 101.4 of MIL-STD-286.

4.4.1.2 Black powder.-The method of determining the percentage of moisture in black powder shall be in accordance with Specification MIL-P-223.

4.4.1.3 Polypropylene felt, paper, and chipboard.-Ten grams of each material shall be accurately weighed and placed In a tared weighing dish. The dish and contents shall be weighed and placed in *an* oven and dried at 100 plus or minus 2 degrees Centigrade for 2 hours. The dish shall be cooled in a disiccator and weighed. The loss in weight shall be calculated as percent moisture in the sample.

4.4.2 Transportation vibration.-The signal assemblies selected as specified in 4.3.3.2.1 and 4.3.3.2.2 shall be packaged and packed in accordance with dwgs. 8836949 and 8836950. Each packed box shall be subjected to the transportation vibration test as specified in MIL-STD-331 except that each box shall be vibrated at the specified amplitudes for four (4) hours in each of three different positions (i.e. box positioned so that signal assemblies are vertical with base end down, box positioned so that signal assemblies are horizontal, and box positioned so that the signal assemblies are vertical with base end up). After the test, the box packing and the signal assemblies shall be examined to determine compliance with the requirements.

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4.4.3 Functioning (Proving Ground).-The signals shall be immersed to a depth between 6 and 9 inches for 2 hours in water maintained at 70 plus or minus 10 degrees Fahrenheit (F.). At the end of the immersion time, the signals shall be removed from the water and the exterior surfaces wiped dry. The signals shall be fired in a pistol, for which the signal is standard, that is mounted on a tower platform at a min. height of 100 feet above the ground. The signal functioning shall be observed for compliance with the requirements. The test shall be conducted in accordance with Acceptance Test Procedure SI-A-VAR, Suppl. SI-A (latest revision).

4.4.3.1 Test validity. -If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.4.4 Air leakage test .-The signal assembly shall be placed in a suitable air tight chamber. The air pressure within the chamber shall be raised to a min. of 3 pounds per square inch and maintained for 15 seconds. An accurate pressure measuring instrument shall be observed for the 15 second interval for evidence of signal leakage.

4.4.5 Static test of star assembly.-The star assemblies shall be tested by supporting the star in a horizontal position with the axis of the star perpendicular to the photometric axis, and ignited by means of quickmatch. The candlepower and color value shall be measured and recorded to determine compliance with the candlepower and color value requirements specified on the applicable drawing. Determination of candlepower and color value shall be made in accordance with the procedures and equipment specified in dwgs. 9201136, 9201268, 92013900 and 9201392.

4.4.6 Chamber gaging.-The signal assembly shall be gaged with equipment specified in 4.3.4.

4.4.7 Jolt test. -The signal assemblies shall be tested in accordance with the procedure specified in MIL-STD-331, Test 101.

5. PREPARATION FOR DELIVERY

5.1 Preparation and packaging.

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5.1.1 Level A.-Signals shall be packaged in accordance with dwg. 8836949.

5.2 Packing

5.2.1 Level A.-Signals shall be packed in accordance with dwg. 8836950.

5.3 Marking.-Marking shall be in accordance with dwg. 8836949 and 8836950.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified In Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

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b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. -Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HBBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. The contractor must include in his submission of designs a listing of any and all Government SIE designs which will be utilized. Partial submission of inspection designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

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6.4 Star burning time.-The star burning time is defined as the time from the appearance of the first colored light from the illuminant assembly (not the first sputtering light from the quickmatch or the black powder priming charge) until the time that the light from the star has decreased to approximately less than 10 percent of the peak brilliance.

6.5 Cost of check test .-The contracting officer should arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The tests shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the primers in the signals and shall not constitute a basis for rejection against either contractor except where deterioration has occured as a direct result of carelessness in handling, storage, etc., permitted while the primer lots were under the jurisdiction of either contractor.

6.6 The signal assembly will be considered safe to transport provided no evidence exists of loose powder or composition in the box, or missing, loose or protruding primers.

6.6.1 The signal assembly will be considered safe to handle provided there is no evidence of leakage of powder or loose composition.

6.7 The signal assemblies will be considered free of damage that will effect the intended function provided the top seal has not been broken by movement or displacement of the top, or the case side has not been distorted sufficiently to prevent ejection of the star assembly.

6.8 Identification of international standardization agreements. Certain provisions of this specification are the subject of international standardization agreement STANAG 3398 Air Pyrotechnics Interchangeability Chart. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

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6.9 Proving Ground test summary:

Lot	Sample Size	Test	Requirements
First Article	30 each designation	Functioning	See Table III
	10 each designation	Jolt	See Table V
Regular Production (First three Consecutive Acceptable Lots)	200	Functioning	See Table III
	32	Jolt	See Table V
Remaining	80	Functioning	See Table III
	20	Jolt	See Table V

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1370-A-014

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Picatinny Arsenal
Dover, New Jersey 07801

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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